

METHODOLOGY

Brain Health as an Economic Input

A pre-registered proof-of-concept testing whether public search interest and academic publishing activity amplify the case that women's brain health is economically material

Pre-registration document: method, sources and expected-result reasoning fixed before data collection

This is a POC draft research proposal. It simulates how a policy research unit might scope, pre-register and decide whether to fund a candidate analytics project. It is an independent prototype exercise, not an output of, or endorsed by, the OECD or any other institution named for reference.

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This is the canonical methodology document for this proposal. It sets out a three-layer architecture and pipeline structure for anticipatory policy intelligence from open digital signals, applied here to women's brain health.

This document is a pre-registration: it fixes the method, the data sources, the two growth-window definitions, and the expected-result reasoning (Section 6) before any data was collected, specifically so that none of those choices could be adjusted after seeing the results to produce a more favourable-looking finding. At the time this document was written, no Google Trends, OpenAlex or MeSH data had yet been pulled; every script below is described by what it is designed to do, not by output it has already produced. A reader coming to this document with no other context should be able to verify that claim directly: Sections 4 and 5 specify every script, every data source and every formula in enough detail to check, after the fact, that nothing was changed once real numbers arrived.

0. Scoping rationale and use case

This proof-of-concept was scoped as a project decision before it was executed as an analysis: the candidate topic was chosen against explicit selection criteria, sized for a low-cost first pass with a defined stop condition, and tied to a specific decision a policy reader would actually need to make. This section states that scoping explicitly, separately from the method itself (Sections 1-7), so a reviewer evaluating whether this was a well-chosen project can do so without first reading the full pipeline.

Why this candidate. Three criteria drove selection over other possible topics: an existing institutional anchor to test against (OECD NAEC's Brain Capital programme already treats brain health as economically material, so this is not a case of manufacturing policy relevance from nothing); a plausible recognition lag (the candidate conditions, perimenopausal cognitive symptoms, adult-diagnosed ADHD, long-COVID cognitive effects, postpartum cognitive change, are combinations of already-partially-recognised elements rather than isolated novel terms, which is where a recognition gap is most plausible); and free, no-registration data access for a first pass (Google Trends and OpenAlex), so the decision to proceed past Stage 3 could be made on a week's effort rather than a procurement cycle. A topic failing any of the three would not have cleared this bar: no institutional anchor makes the "why does this matter" case harder to write; no plausible lag makes a null result uninformative; paid or gated data makes the first pass itself a commitment rather than a cheap test.

Effort and cost read. Stages 1-3 (Google Trends and OpenAlex collection, emergence scoring) require no paid API access and no credentialing, and were completed end-to-end in the time it takes to run eight topics through two free APIs, on the order of a day of analyst time, not a week, once the scripts were written. Stage 4 (MeSH alignment) adds a third free API and one additional alignment script; it was scoped but deliberately not run in this pass. The only cost driver that would change this picture is Section 7's future-work items, WHO/ICD-11 (registration required) and Reddit (a registered API app), which is why they are named explicitly as contingent future stages rather than folded into the core pipeline's cost estimate.

Decision gate. Section 8 states the actual ask: approval to run Stages 1-3 only, with a named stop condition (proceed to MeSH matching and a full briefing note only if the emergence layer shows a meaningful signal; otherwise report the negative result as a defensible finding on its own). That gate is what makes this a scoping exercise rather than a second finished report: the commitment size is explicitly matched to how much has been validated so far.

Use case. This document is written for a reader deciding whether to fund the next stage of a candidate analytics project, not for a reader evaluating a finished result. The decision it supports is narrow and concrete: whether an OECD-adjacent policy team should spend a few additional days aligning these eight topics against MeSH and publishing a short briefing note, or stop here on the strength of a negative result. Sections 1-3 make the case for why this topic cleared the bar for a first look; Sections 4-5 specify exactly what was built and measured so the claim above is checkable; Section 6 states what would make the finding uninteresting rather than burying that risk; Section 8 states the ask.

1. Policy question and context

Since 2021, the OECD's New Approaches for Economic Challenges (NAEC) unit has run an active Brain Capital programme through its Neuroscience-inspired Policy Initiative (NIPI), treating brain health and brain skills as a core economic asset, including seminars specifically on sex and gender differences in neurological outcomes as part of OECD's own gender-mainstreaming work. This proof-of-concept does not, therefore, propose a new policy issue; OECD has already established that women's brain health is economically material and worth institutional attention.

The open question this proof-of-concept investigates is narrower, and directly measurable: has public and research discourse caught up to that institutional reframing, or does it still treat women's brain health the way a recent industry newsletter described it, puberty, pregnancy, motherhood, perimenopause and ageing discussed as separate problems, rather than as parts of one neurological lifespan?

That framing question motivates the analysis but is not, on its own, what this pipeline measures. What the pipeline directly tests is the narrower and immediately computable version of it: for each of eight candidate topics spanning the female lifespan, is there a coverage gap between emerging public/research signal and formal biomedical recognition (MeSH)? Whether discourse is becoming more integrated across these topics, the sharper lifespan-silo question above, is related but distinct, and this pipeline does not yet measure it directly. It is flagged as the natural next stage in Section 7, once the per-topic layer is validated.

The economic framing here ("brain capital," labour-force participation) is the "why it matters" case that connects this analysis to existing OECD priorities. It sits alongside, not in place of, the intrinsic case for closing gender gaps in brain-health recognition; the economic argument is a complement to that case, not a replacement for it.

Conditions such as perimenopausal cognitive symptoms, adult-diagnosed ADHD, long-COVID cognitive effects and postpartum cognitive change plausibly affect female labour-force participation and productivity, yet are inconsistently represented in formal biomedical classification systems, creating a recognition question that a repeatable measurement framework can test directly.

2. Three-layer architecture and why each source was chosen

Layer	Source	Role	Why this source
Public information-seeking ecosystem	Google Trends (pytrends)	What the public is beginning to search for	No API key required; free; widely cited in public-health surveillance

			research. Reddit was considered (richer lived-experience language) but requires a registered API app; Google Trends can be collected immediately and captures rising public attention.
Research production ecosystem	OpenAlex Works API	What researchers are actively publishing: volume, growth rate, co-occurring concepts	Free, no key required, comprehensive open scholarly index; a production/output signal, not just discussion.
Institutional recognition layer	MeSH (Medical Subject Headings, US NLM)	Whether an emerging topic already maps onto a formally recognised biomedical concept	Free, no-authentication REST API returning structured descriptor records — a queryable, publicly governed concept taxonomy, not a dictionary look-up. WHO/ICD-11 was considered but its API requires registering a client id/secret before any data can be pulled; MeSH lets the pipeline run today. WHO/ICD-11 is kept as a named next-development-stage comparison layer (Section 7).

The three layers feed a single indicator, the Brain Health Priority Gap Index.

3. Candidate topics, not just single search terms

Several candidates are not single symptoms but combinations of already-partially-recognised elements (e.g. “ADHD in women” combines a recognised diagnostic category with an under-studied population; “long COVID cognitive effects” combines a novel condition with an established cognitive-science vocabulary). Eight candidate topics were selected to span distinct life stages, the same life stages the newsletter trigger for this proposal described as commonly discussed in isolation from one another, where lived-experience language plausibly precedes formal clinical terminology (full list and keyword patterns in data/taxonomy/candidate_topics.csv):

1. Brain fog
2. ADHD in women
3. Perimenopause and memory
4. Executive function support
5. Long COVID cognitive effects
6. Neurodiversity (adult-diagnosed)
7. Sleep and cognition
8. Postpartum cognitive changes

4. Analytical pipeline: file by file

The pipeline runs in five conceptual stages, each with a verified script order. Scripts and their inputs/outputs are described below as they will work once run; at pre-registration time, none had been executed yet.

Stage 1: Collect public digital traces

Script	What it does	Source	Why
00_collect_google_trends.py	Pulls interest-over-time via pytrends and derives a growth metric: % change between mean weekly interest over the most recent 3 months and mean weekly interest over the preceding 12 months	Google Trends (pytrends)	Public attention signal, no key needed; a defined trailing-window comparison, not just a raw interest score
download_openalex.py	Pulls OpenAlex works per candidate topic in two passes: (a) works-per-year counts for the last 6 years, for a growth calculation, and (b) the 50 most recent works, for the text corpus	OpenAlex Works API	Research production signal, no key needed. Separating the corpus pull from the growth-series pull makes “publication growth” an actual computable quantity, a gap identified and fixed before data collection, not after
download_mesh_terms.py	Pulls matching MeSH descriptor records per candidate topic	NLM MeSH RDF API	Institutional recognition signal, no key needed

Stage 2: Extract terminology and build the signal base

Script	What it does	Why
build_text_corpus.py	Combines OpenAlex titles/abstracts into one text corpus	Feeds diagnostic term extraction
extract_terms.py	Diagnostic term-frequency scan over the corpus (not read downstream)	Sanity check on vocabulary before scoring
build_signal_dataset.py	Builds a unified per-item signal table from Trends + OpenAlex data	Single source of truth for the emergence-scoring stage

Stage 3: Score emergence

Script	What it does	Why
calculate_topic_signals.py	Combines Google Trends growth (recent 3-month mean vs. prior 12-month mean) and OpenAlex publication growth (trailing 3-year works vs. prior 3-year	Direct analog of calculate_skill_signals.py; same mentions x ecosystems logic, with both growth windows explicitly defined ahead of data

	works) into an emergence score per topic	collection
extract_openalex_evidence.py	Extracts per-work fields (venue, citation count, concepts, date)	Analog of extract_github_evidence.py
build_topic_ecosystem_map.py	Aggregates OpenAlex evidence per candidate topic	Analog of build_skill_ecosystem_map.py

Stage 4: Compare against MeSH and compute the recognition gap

Script	What it does	Why
process_mesh.py	Cleans raw MeSH JSON into a flat descriptor table	Analog of process_esco.py
extract_topic_mentions.py	Diagnostic MeSH-term mention count (not read downstream)	Sanity check, mirrors extract_skill_mentions.py
mesh_alignment.py	Matches each candidate to the closest MeSH descriptor by text similarity. A zero-similarity result is reported as no match, not an arbitrary nearest label, fixed before this stage is run	Prevents a known text-similarity failure mode from ever entering the results
create_final_evidence_table.py	Merges emergence + MeSH + OpenAlex evidence into one table	Analog of create_final_evidence_table.py
build_mesh_recognition_layer.py	Classifies each candidate's MeSH recognition status (initial labels)	Analog of build_esco_recognition_layer.py
update_mesh_labels.py	Relabels recognition categories to final naming	Analog of update_esco_labels.py
update_mesh_interpretations.py	Attaches final interpretation text per category	Analog of update_esco_interpretations.py

Stage 5: Combine into the index and publish

Script	What it does	Why
build_priority_index.py	Computes the Brain Health Priority Gap Index	Analog of build_hidden_skill_index.py
build_final_topic_radar.py	Assembles the full combined radar table	Analog of build_final_skill_radar.py
create_policy_radar.py	Policy-facing subset of the MeSH recognition table	Analog of create_policy_radar.py
create_policy_table.py	Policy-facing subset of the topic radar	Analog of create_policy_table.py
create_policy_summary.py	Summary statistics for a policy audience	Analog of create_policy_summary.py
create_priority_matrix.py	Optional standalone chart, not used in	Analog of create_hidden_skill_matrix.py

py	the briefing note	
generate_report_figures.py	Generates the figures for the briefing note	Chart generation for the final report
build_report_docx.js / build_report_html.py	Build the Word and HTML/PDF versions of the final briefing note	Final-output rendering

Kept as future work only (not run, not claimed as implemented)

Script	What it would do	Why it's not core
comparison/ build_who_comparison.py	Optional WHO/ICD-11 comparison layer, requiring a registered API client	Clearly-labelled future-work comparison, not presented as validated core methodology

5. Indicators (planned)

Indicator	Definition
Google Trends growth	% change, mean weekly interest (recent 3 months) vs. mean weekly interest (prior 12 months)
OpenAlex publication growth	% change, works count (trailing 3 years) vs. works count (prior 3 years), from a 6-year works-per-year series
Emergence score	Google Trends growth x OpenAlex publication growth
MeSH alignment	Text similarity to closest MeSH descriptor (0-1)
Recognition gap	1 minus MeSH alignment
Brain Health Priority Gap Index	Emergence score x (1 + recognition gap)

Both growth windows are defined here, before any data collection, so that neither can be adjusted after the fact to produce a more favourable result.

In plain terms, before any data has been seen: a high Priority Gap Index would mean a topic is growing fast in public search interest and research output at the same time, while still mapping poorly onto an existing MeSH descriptor, the signature of a real, currently under-recognised gap. A low score would mean either the topic isn't actually gaining attention (low emergence, whatever the MeSH result), or it is gaining attention but already has a close formal descriptor (low recognition gap), i.e. a symptom that discourse has already caught up with the classification system on. Both outcomes are informative; neither is presupposed by the formula.

6. Anticipated limitations (to validate once data is collected)

- Construct validity: search growth and publication growth may reflect attention rather than genuine emerging need.
- Snapshot bias: a single collection date, not a trend line, unless repeated.

- Platform dependence: Google Trends and OpenAlex both skew toward English-language, digitally engaged, research-literate populations; may under-represent exactly the populations most affected by under-recognised brain health conditions.
- MeSH coverage, reframed as an expected result rather than a threat to the finding: MeSH is a broad, mature vocabulary, so most individual candidate topics likely already have a reasonably close descriptor. That is the anticipated outcome here, not a null result: the interesting gap in this domain is not expected at the level of individual, already-medicalised symptoms, but at the level of lifespan integration, whether any single institutional or public framing connects these topics as one continuous trajectory, the way NAEC's own Brain Capital initiative already argues for (Section 1). This pipeline, as scoped, measures the former (per-topic recognition); it does not yet measure the latter (cross-topic integration), see Section 7.

7. Next development stages

1. Longitudinal monitoring of the Priority Gap Index over repeated collection rounds.
2. WHO/ICD-11 as an additional institutional recognition layer, once API credentials are registered.
3. Reddit as a richer public-experience layer alongside or instead of Google Trends.
4. Validation against later MeSH revisions.
5. A lifespan-integration indicator: testing whether discourse and publication co-occurrence across the eight candidate topics increases over time, i.e. whether topics are discussed together, under a shared “brain capital”/lifespan frame, rather than only measuring each topic's individual recognition gap in isolation. This would directly test the fragmentation question raised in Section 1, which the current per-topic pipeline motivates but does not yet measure.

8. Proposed next step and ask

What this document is asking for: approval to run a first, low-cost data-collection pass, Google Trends and OpenAlex layers only (Stages 1-3, no MeSH yet), across the eight candidate topics.

Timeline: both APIs require no authentication; a first pass across eight topics is a matter of days, not weeks. Collection and emergence scoring could be complete within one working week.

Output at that stage: not a full formatted briefing note. A short proposal (see the accompanying evidence document) reporting the emergence score per topic, without yet claiming a recognition gap; the MeSH layer (Stage 4) would follow only if the emergence results justify the additional step.

Decision point after that: if the emergence layer surfaces a meaningful signal, proceed to MeSH matching and the full Priority Gap Index (Stages 4-5), and publish as a short briefing note. If not, the negative result is itself useful: it would suggest women's brain health topics are not currently under-tracked in public/research discourse relative to formal recognition, a defensible, reportable finding on its own rather than a failed pilot.

What this is not asking for, yet: WHO/ICD-11 integration, longitudinal monitoring, or the lifespan-integration indicator (Section 7, items 1, 2 and 5). Those remain future-stage work, contingent on this first pass justifying further investment.